13 Ch 13 – Business Valuation

Asked in last 3 papers: Sep 2025 Q1(a) 6m (ke by CAPM, EPS, equity value from EBITDA multiple, EV = market cap + debt – cash); Jan 2026 Q4(b) 4m (EVA with CAPM ke, EVA dividend per share); May 2026 – nil.

1. Conceptual Framework of Valuation

→ Valuation = assessing the **'true worth'** of an enterprise; both science and art; part of strategic financial management. Used here mainly for **M&A**.

→ Need for valuation: (a) information for **internal stakeholders**; (b) **comparison** with similar enterprises (management efficiency); (c) future **public listing**; (d) **strategic planning** (value driver, deployment of surplus cash); (e) **ball-park price** for acquisition.

2. Important Terms (2.1–2.5)

Term	Meaning / logic	SM figures
2.1 PV of cash flows	Time value of money: ₹ received later is worth less today.	₹ 1,000 a year hence at 10% ≈ **₹ 909**
2.2 IRR	Discount rate at which NPV of all cash flows = 0 (PV of inflows = investment). Higher IRR → more likely selected.	—
2.3 ROI	ROI = Return ÷ Investment. With additions during the year, use **average investment** (recognises timing). **Historical** ratio → only course correction / channel more money to the better business line.	(1400 – 1000)/1000 = **40%**. Opening 1000 + 200 added, closing 2000: return 800 ÷ avg (1000+1200)/2 = 1100 → **72%**
2.4 Perpetual growth (Gordon)	Assumes perpetual growth in dividend. Not widely used: (i) more parameters to factor in, (ii) dividends rarely grow at a steady rate. "Darling of academicians" (constant-rate model).	—
2.5 Terminal Value (TV)	Value at exit point = **'Horizon Value'**; mostly by Gordon perpetual growth: TV = (CF at Year 3 × growth rate) ÷ (WACC – growth rate), SM computes it as (517.29 × 1.03)/(0.08 – 0.03), i.e. CF × (1 + growth rate).	see 3.3

3. Approaches / Methods of Valuation

→ Unlisted target → price = **negotiated**. Listed target → market cap is a guide but not the worth (depends on **market efficiency**; often only a small portion of shares is quoted).

→ Three broad approaches: (a) **Asset based**, (b) **Earning based**, (c) **Cash flow based**; plus other methods.

3.1 Asset Based Approach (3.1.1–3.1.3)

→ Value of shares = **net assets acquired**; based on income-generating assets, not income. **Least important for IT companies** (assets are IPR and human resources).

Method	Basis / formula	Plus / minus
3.1.1 **Net Asset Value** (Book Value)	Net assets = Fixed Assets + Net Current Assets – Long Term Debt (historical cost)	Simplest, data easily available; but Balance Sheet is not a valuation device → gives the **lower limit**; ignores current values, intangibles (brand, IPR).
3.1.2 **Net Realizable Value** (Liquidation / Adjusted Book Value)	Realisable value of all assets – liquidation expenses – liabilities (liquidation expenses may be ignored if acquired as going concern)	Off-market purchase → buyer may offer lowest price; not popular (total break-up); useful where acquirer sells one part and integrates the rest.
3.1.3 **Replaceable Value**	Cost of an equivalent group of assets and liabilities in the open market	Better than book value; slightly **higher than NRV** (quick disposal not encouraged) → many treat it as the **maximum price** an acquirer would pay; cannot value staff loyalty.

Note (app-written): the SM prints the 3.1.1 left-hand side as "Net Fixed Asset"; the result is the net asset (book) value of the shares.

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SM example (NRV): *Liabilities 20,000 (both bases). Non-current: PPE 50,000 → 40,000, Licences 10,000 → 30,000 (60,000 → 70,000). Current: Debtors 50,000 → 45,000, Cash 10,000 (60,000 → 55,000). Net assets book **1,00,000** vs NRV **1,05,000**; 5,000 shares → **₹ 21** per share.*

→ ***Conclusion:*** *depicts net worth fairly using 'going concern'; major drawback — ignores ability to generate **future revenues** and market dynamics.*

3.2 Income Based Approach (3.2.1–3.2.2)

→ *Uses **earning potential**; suits an acquirer who will **continue** the target's business. Additional earnings due to acquisition (synergy) are included. Two versions: **PE ratio / earning yield** and **capitalisation rate**.*

Formula	Terms / logic	SM use
3.2.1 Price per share = **EPS × PE Ratio**	For listed cos (PE available); takes in expected growth and market expectations. Unlisted → PE of equivalent quoted co / industry. Gives the **minimum acceptable price** to target's shareholders.	TYK P2, P12
Required earning yield = EPS ÷ Share price = 1 ÷ PE	Approximates the capitalisation rate.	—
3.2.2 Capitalised Earning Value = Expected Annual Maintainable Profit ÷ Capitalisation rate	Maintainable profit = **weighted average** of past profits adjusted for synergy / economies of scale.	TYK P2, P4; Illus 2

→ *PE method steps: (i) choose PE of **equivalent quoted company**; (ii) adjust it **downward** for non-listing risk; (iii) find **future maintainable EPS**; (iv) EPS × adjusted PE.*

→ *Capitalisation: **forward looking** (advantage); estimating future profit and treatment of **extraordinary / exceptional** items (disadvantages).*

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TYK P2 (Eagle Ltd): *PAT 77 ÷ (1 − 0.30) = 110 − extraordinary income 8 + extraordinary loss 10 = 112; new product 70 − 20 − 12 − 10 = 28 → 140; tax 42 → maintainable PAT **98**; ÷ 0.14 → value of business **₹ 700 lakhs**. Pref. dividend 13% × ₹ 1 cr = 13 → equity earnings ₹ 85,00,000 ÷ 50,00,000 = EPS **₹ 1.70** × PE 10 = **₹ 17**.*

TYK P4 (H Ltd / B Ltd): *NAV: H (1,300 − contingent liability 300) ÷ 3.50 cr = **₹ 285.71**; B 31.50 ÷ 0.65 = **₹ 48.46**. Earnings capitalisation at 8%: H 300/0.08 ÷ 3.50 = **₹ 1,071.43**; B 10/0.08 ÷ 0.65 = **₹ 192.31**. Fair value (weights 1 : 3): H **₹ 875**, B **₹ 156.3475** → exchange ratio **0.1787** H share per B share (contingent liability assumed to materialise in full).*

3.3 Cash Flow Based Approach (DCF)

→ *Takes the **free cash** available in future periods and discounts it at a rate matching the flow's risk; overcomes over-reliance on **historical data**. PV > current cost of investment → valuation attractive.*

→ *Five steps: (a) arrive at **Free Cash Flows**; (b) **forecast** future cash flows; (c) **discount rate** from cost of capital; (d) **Terminal Value**; (e) PV of FCFs and TV, and interpret.*

→ *Limitation: result **gravitates heavily on WACC and TV**; TV is often a significant portion → growth rate and assumptions must be validated.*

Formula	Terms / logic	SM use
FCF = EAT − one-time incomes + one-time expenses + depreciation	One-time events eliminated; depreciation is a book entry.	SM example
Adjusted FCF = FCF − change in working capital − investment in fixed assets	Reinvestment needed to earn the flow.	SM example
Free Cash Flow (FCF) = EAT − (Capital Expenditure − Depreciation) − Δ Working Capital	EAT = EBIT after tax (no interest in FCFF); net capex and extra working capital are cash out.	TYK P3, P7, P8
Terminal Value = FCF of terminal year ÷ (Cost of capital − growth rate); PV = TV × PVF of last explicit year	App-written generalisation of the TYK P7/P8 working: e.g. 375.32/(0.12 − 0.10); cost of capital of the stable phase.	TYK P7, P8
$V_0 = FCFF_1 \div (K_c - g_n)$	$FCFF_1$ = next year's FCFF; K_c = cost of capital; g_n = growth forever.	TYK P6

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SM example (₹ '000): EAT 600 − 200 + 100 + 100 = FCF **600**; growing 5%: 600, 630, 661.50; less ΔWC (50, 30, −10) and FA (50, 50, 20) → adjusted FCF **500, 550, 651.50**; WACC 8%, PVF 0.926, 0.857, 0.794 → PV **463.00, 471.35, 517.29**; TV at 3%: (517.29 × 1.03)/(0.08 − 0.03) = **10,656.17**; total DCF = **12,107.81** (≈ 12,108 thousand).

Note (app-written): the SM's TV is built on the already discounted year-3 flow (517.29), so 10,656.17 is already a present value (= 651.50 × 1.03 ÷ 0.05 × 0.794); that is why it is added without further discounting.

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TYK P1 (ABC / XYZ): market value 400 × 1.5 cr = **₹ 600 cr**; DCF 250 × 0.893 + 300 × 0.797 + 400 × 0.712 = **₹ 747.15 cr** (₹ 498.10/share) → range: minimum ₹ 400 (600 cr), maximum **₹ 498.10** (747.15 cr).

TYK P3 (ABC Co, new strategy): sales +20% for 3 yrs: 24,000 / 28,800 / 34,560 / 34,560; FA 40%, CA 20% of sales; PAT 1,680 / 2,016 / 2,419.20 / 2,419.20; dep 10% of opening FA 800 / 960 / 1,152 / 1,382 → FCFF **(720), (864), (1,036.80), 2,419.20**; PV at 15% = **−1,961.79**; residual 2,419.20/0.15 = 16,128 ÷ 1.521 = **10,603.55**; value 8,641.76 − pre-strategy 1,400/0.15 = 9,333.33 → **−691.57** → not viable.

TYK P5 (AB / XY Ltd): AB price (570 + 430)/2 = **₹ 500**. DCF at 8%: 105 × 0.93 + 120 × 0.86 + 125 × 0.79 + 120 × 0.74 + (100 + TV 200) × 0.68 = **₹ 592.40 lakhs**; NAV (150 + 200 − loans 100) = **250**; average **421.20** → 4,21,20,000/500 = **84,240** shares; fully-paid equivalent 20 + 5 = 25 lakh → fully paid 84,240 × 20/25 = **67,392**, partly paid **16,848**.

TYK P6 (Hansel): 1,800 = 54/(K_c − 0.09) → K_c 12%; 20%(1 − X) + 10%X = 12% → book debt weight 80%. Market: equity 20 × 3 = 60, debt 80 × 0.9 = 72 → K_c = 20%(60/132) + 10%(72/132) = **14.5455%**; value = 54/(0.1454 − 0.09) = **₹ 974.73 lakhs**.

TYK P7 (XYZ Ltd, two-stage): high growth ke 10% + 1.15 × 6% = 16.9%, kd 13%(0.7) = 9.1% → WACC **13%**; stable ke 9% + 1 × 5% = 14%, kd 9% → 0.6 × 14 + 0.4 × 9 = **12%**. FCF = EAT − (capex − dep, growing 20% from 80) − ΔWC (25% of revenue increase): **56.00, 67.20, 80.64, 96.77**; terminal year 479.00 − 103.68 = **375.32**. PV @13% = **217.38**; TV 375.32/(0.12 − 0.10) = **18,766** × 0.613 = **11,503.56** → firm **₹ 11,720.94 cr**.

TYK P8 (WXY Ltd): revenue 9,000 / 10,800 / 12,960 / terminal 13,996.80; EBIT 2,700 / 3,240 / 3,888 / 4,199.04; EAT 1,890 / 2,268 / 2,721.60 / 2,939.33; capex − dep (150 × 1.15^n) 172.50 / 198.38 / 228.13 / nil; ΔWC 375 / 450 / 540 / 259.20 → FCFF **1,342.50, 1,619.62, 1,953.47**, terminal **2,680.13**. PV @15% = **3,676.44**; TV 2,680.13/(0.15 − 0.08) = 38,287.57 × 0.6575 = **25,174.08** → firm **₹ 28,850.52 cr**.

Note (app-written): TYK P5 prints "(120x0.74)x(300x0.68)"; the last "x" should be "+" (592.40 is the sum; 300 = year-5 earnings 100 + TV 200), SM p.58. TYK P6: ICAI rounds K_c to 0.1454 in the denominator; with 14.5455% unrounded the value is ≈ ₹ 973.77 lakhs (ICAI's ₹ 974.73 kept), SM p.60.

4. Measuring Cost of Equity — 4.1 CAPM, 4.2 APT

Asked: Sep 2025 Q1(a) 6m

Formula	Terms / logic	SM use
4.1 $R = rf + \beta (rm - rf)$	R = expected rate of return (cost of equity); rf = risk free rate of return; β = Beta value of the stock; rm = market rate of return. Returns are a reward for assuming risk; two buckets: risk-free + market-relative risk.	Illus 1, 2; TYK P7, P10, P16
4.2 $Rf + \beta1(RP1) + \beta2(RP2) + ...\beta j(RPn)$	Each macro-economic factor (e.g. interest rate, sector growth) has its own beta; RP = risk premium of the factor. E(R) = **linear function** of factors.	—

→ CAPM's weakness: **oversimplification of risk**. APT (Stephen Ross, 1970s) uses **multiple factors**; APT itself does not name the factors — analysts must develop them for their economy.

4.3 Estimating Beta and Valuation of Unlisted Companies

→ Biggest challenge for a private firm: **cost of capital** → **beta**. Listed peers are leveraged; private firms have nil/insignificant debt → use **unlevered beta**.

→ Same problem for a listed company entering a **new line of business**: do not use own WACC; use **asset (ungeared) beta** re-geared to own gearing. Asset beta = only **business (systematic) risk**, no financial risk.

→ Equity beta is always > asset beta; debt-free → equal. Debt beta assumed **zero** (bond returns not linked to market volatility).

Formula	Terms / logic
$\beta_a = \beta_e \times E/[E + D(1-t)] + \beta_d \times D(1-t)/[E + D(1-t)]$	β_a asset (ungeared) beta; β_e equity (geared) beta; β_d debt beta; E equity; D debt; t tax rate. Weighted like WACC.
$\beta_d = 0$: $\beta_a = \beta_e \times E/[E + D(1-t)]$	De-gear a proxy's beta.
Re-gear: $\beta_e = \beta_a \times [E + D(1-t)]/E$	Uses the **appraising company's** own capital structure. App-written simplification: $= \beta_a[1 + (1-t)D/E]$, the form used in Illus 1.

→ Steps: (1) identify **pure-play / proxy** firms and their equity betas; (2) **de-gear** to asset beta; (3) one proxy → use it; more → **average** (or the most appropriate); (4) **re-gear** at own D/E; (5) put β_e in **CAPM** → required return / value.

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Illus 1 (X Pvt Ltd): levered β = 1.8[1 + (1 − 0.3) × 40/60] = **2.64**. Adjusted EBITDA 90 − extraordinary gain 10 − pending write-off 20 = **60** × 5 = EV **300**. ke = 5% + 2.64(11% − 5%) = **20.84%**; WACC = 20.84 × 60/100 + 12 × (1 − 0.3) × 40/100 = **15.864%**. FCF 100, 120, 150 × DF 0.863, 0.745, 0.643 = 86.30 + 89.40 + 96.45 → value of the firm **272.15**.

5. Relative Valuation

Asked: Sep 2025 Q1(a) 6m

→ Asset, earnings and cash-flow approaches give **intrinsic value**. Relative valuation gives a **'relative' value** by **comparative analysis with peers** (similar asset and risk dispositions); increasingly used **with** the intrinsic approaches to validate them. Also called **'valuation by multiples'**. (TYK T2 → this para.)

→ Steps: (1) find the **drivers** for the multiple; (2) determine results through **financial ratios**; (3) find **comparable firms** and compare; (4) **iterate** to smoothen deviations.

Multiple	Best suited (SM)
EV/EBITDA	Popular; gives critical inputs.
EV/Invested Capital	**Capital-intensive** enterprises; misfit for asset-light firms or those with significant investment properties.
EV/Sales	**Cash-rich**, huge order book, organic growth through own capital.
P/E (equity based)	"Celebrated status"; may not reflect forex risk of an exporter (Brexit example).
PEG = PE Ratio ÷ Growth Rate	**High growth or sunrise** industries.

→ **Step 2 (ratio):** a factor-based approach helps choose the right ratio. **Step 3 (comparables)** — most challenging: same size, capital structure, head count can still differ in revenue model (X: **FTE pricing** vs Y: **Unit Transfer Pricing**) → "take benchmarks with a pinch of salt". Niche market → look at similar companies in other industries; or take closely relevant firms and run a **regression** (last resort).

→ **Assumptions:** (i) market is **efficient**; (ii) function between fundamentals and multiples is **linear**; (iii) comparables similar in **structure, risk and growth**.

→ EV two ways: (a) take entity value and **adjust for debt**; (b) **balance sheet** approach.

Formula	Terms / logic	SM figures
EV = EBITDA × EBITDA multiple	Value of the whole business (debt + equity holders).	Illus 1, 2
Equity value = EV − Debt + Surplus funds (cash)	Debt belongs to lenders; surplus cash belongs to shareholders.	Illus 2: 3,000 − 300 + 500 = 3,200
EV = Market capitalisation + Debt − Cash & cash equivalents	Market cap = shares × CMP (balance sheet approach).	Illus 3: 8,40,000 + 2,00,000 − 5,00,000

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Illus 2 (A Ltd): ke = 4.5% + 0.9(12% − 4.5%) = **11.25%** (capitalisation rate). EBITDA = GP 1,000 − indirect exp. 400 = **600** (₹ '000); earning value 600/0.1125 = 5,333.33 ÷ 1,00,000 shares = **₹ 53.33** per share. (b) 600 × 5 = 3,000 − debt 300 + surplus 500 = equity value **₹ 3,200 thousand**.

Illus 3 (H K Ltd): 70,000 shares × CMP ₹ 12 = market cap ₹ 8,40,000 + long-term debt ₹ 2,00,000 − cash ₹ 5,00,000 = EV **₹ 5,40,000**.

6. Other Approaches — 6.1 Contemporary Approaches to Valuation

→ Technology-driven / conglomerate businesses → new terms: **PEs, Exit Multiples, LBOs, Brand Value**; some methods born of industry peculiarities.

→ Internet company (near-zero fixed assets, high brand recall) → **Price per Page visited**; online play store → **Price per Subscriber**. The discerning still check cash to sales and apply DCF.

→ **Goodwill-based:** asset-based value first, then goodwill separately via a multiple of annual sales or footfall (retail giant buying a mom-pop store).

→ **PER** = Price ÷ EPS (market prices the stock on fundamentals). EPS ₹ 40, avg price ₹ 50 → **1.25**. PER is relative; compare within the sector for a median PER.

→ **LBO:** Private Equity firms buy enterprises "in the rough" with debt and equity (typically **70 : 30** debt to equity), sell in 3–5 years (spin-off control or outright sale). Prime casualty: future can't be predicted — a cyclical slowdown leaves the buyer with huge loan and interest.

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SM example (LBO, ₹ lacs): EBITDA 100; consideration 250 (100% premium over BV + goodwill 50); bank 200, VC 50. EBIDTA Y0–Y3 100, 120, 144, 178; interest 36, 30, 24, 18 (debt repaid linearly over 6 yrs); tax 30% → EBDAT 44.80, 63, 84, **112**. Exit 7 × 112 = **784** − debt 100 = equity value **684**.

Note (app-written): Y3 EBIDTA is printed 178.00 though 20% growth on 144 gives 172.80; the rest of the column (EBDTA 160, tax 48, EBDAT 112) follows 178 (SM p.21). ICAI's figures kept.

6.2 Chop-Shop Method (Break-up value)

- Identifies **multi-industry** companies that are **undervalued** and worth more separated; buy assets **below replacement value**.
- Steps: (1) identify segments and **average capitalisation ratios** of firms in those industries; (2) compute a **"theoretical" market value** from each ratio; (3) **average** them = chop-shop value.

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Illus 4 (Cornett GMBH, market €4 million): Cap/Sales: 0.75 × 1.5 mn + 1.10 × 0.8 mn + 1.00 × 2.0 mn = **€4,005,000**. Cap/Assets: 0.60 × 750,000 + 0.90 × 700,000 + 0.60 × 3,000,000 = **€2,880,000**. Cap/Operating income: 10 × 100,000 + 7 × 150,000 + 6 × 600,000 = **€5,650,000**. Average = 4,178,333.33 → **€4,178,000**.

Note (app-written): the 2nd and 3rd solution tables are headed "Capital-to-Sales / Segment Sales" but use capitalisation/assets with segment assets and capitalisation/operating income with segment income (SM p.22).

6.3 Economic Value Added (EVA)

Asked: Jan 2026 Q4(b) 4m

- Holistic measure of financial performance and economic contribution. A company creates value only if **returns exceed its cost of capital**. EVA **separates operations from financing**. Negative EVA → not generating value; positive → producing value.
- **Why EVA:** ROI, ROCE, asset/earnings/FCF methods are "number-crunching"; EVA makes **management accountable** for decisions (e.g. choosing only the earlier break-even project although both had positive IRR) and shows whether returns **cover the cost of capital**.

Formula	Terms / logic	SM use
EVA = NOPAT − (Invested Capital × WACC) = NOPAT − Capital Charge	Excess of returns over the weighted average cost of invested capital.	All EVA items
NOPAT = EBIT − tax (= EBIT(1 − t)) + non-cash expenses	Operating cash return after tax; depreciation left untouched (operating expense); add back book entries such as **provision for bad and doubtful debts**.	Illus 5
Invested Capital = Total assets − non-interest-bearing current liabilities (creditors, billing in advance) + non-cash adjustments	Also = **equity + long-term debt**, generally at the **start of the year**. Adjust non-cash items in both NOPAT and capital.	Illus 5, TYK P14
Capital charge = Invested Capital × WACC; WACC (ko) = ke × E/(E+D) + kd × D/(E+D), kd post-tax = rate × (1 − t)	Hurdle the operations must clear.	TYK P9–P16
Taxable income = Net income ÷ (1 − t); EBIT = Taxable income + Interest	Back out operating income from PAT (used in TYK answers).	TYK P10, P11, P13
Financial leverage = PBIT ÷ PBT = PBIT ÷ (PBIT − Interest)	Gives PBIT when only leverage and interest are known.	TYK P9
EVA dividend per share = EVA ÷ No. of shares	Maximum DPS before the value of the entity starts declining.	TYK P11, P13

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Illus 5 (A Ltd, ₹ lac): NOPAT = EBIT 410 − tax 123 + bad-debt provision 20 = **307**. Invested capital = 1,300 − 400 = 900 + 20 = **920** (current liabilities include 100 tax liability). WACC = 800/900 × 8.45% + 100/900 × 12%(1 − 0.30) = **8.44%**. Capital charge 920 × 8.44% = **77.65** → EVA = 307 − 77.65 = **229.35**.

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TYK P9 (Jatayu): interest 10% × 400 = 40; 1.5 = PBIT/(PBIT − 40) → PBIT **₹ 120 lakhs**; NOPAT 120 × 0.7 = **84**; WACC = 14% × 300/700 + 7% × 400/700 = **10%**; EVA = 84 − 0.10 × 700 = **₹ 14 lakhs**.

TYK P10 (RST): ke = 8.5% + 1.36 × 9% = **20.74%**; kd = 11% × 0.7 = 7.70%; WACC = 20.74 × 125/165 + 7.70 × 40/165 = 15.71 + 1.87 = **17.58%**. Taxable 25,00,000/0.7 = 35,71,429 + interest 4,40,000 = EBIT **40,11,429**; EVA = 28,08,000 − 29,00,700 = **−₹ 92,700**.

TYK P11 (Tender): 15 lac/0.7 = 21,42,857 + interest 10,00,000 = operating income **₹ 31,42,857**; EVA = 22,00,000 − 13% × 95,00,000 (12,35,000) = **₹ 9,65,000**; ÷ 6,00,000 shares = **₹ 1.6083** per share.

TYK P12 (Orange / Grape / Apple): kd post-tax 10.40% / 8.45% / 9.75%; WACC **13.52% / 15.225% / 17.95%**; EVA = 25,000 × 0.65 − WACC × 1,00,000 = **2,730 / 1,025 / −1,700** → **Orange** best (highest EVA, lowest WACC). Interest 12,800 / 6,500 / 3,000 → NI 7,930 / 12,025 / 14,300 → EPS 1.30 / 1.45 / 1.43 × PE 11 = **₹ 14.30 / 15.95 / 15.73**; market cap **87,230 / 1,32,385 / 1,57,300**. PE should be adjusted for different financial risk.

TYK P13 (Delta): 15,00,000/0.6 = 25,00,000 + interest 15,00,000 = EBIT **₹ 40,00,000**; EVA = 24,00,000 − 0.126 × 1,00,00,000 = **₹ 11,40,000**; ÷ 2,50,000 = **₹ 4.56**/share. No dividend → value expected to **increase** (higher growth, higher EBIT).

TYK P14 (XYZ Inc, $ mn): NOPAT 180 − 63 = 117 − 12% × capital employed 605 (100 + 325 + 180) = 72.60 → EVA **44.40**. MVA = market value of equity 10 mn × $50 = 500 − equity fund 425 = **75**.

TYK P15 (Herbal Gyan): investment = WC 20 + PPE 80 + patent (offer value) 40 = 140; EVA = 12 − 0.15 × 140 = **−₹ 9 lakh**.

TYK P16 (Constant Engineering): β = 1 (moves with market) → ke = 10% + 1(12% − 10%) = 12%; investment = NW 200 + LTD 400 + patent 100 = 700; WACC = 12 × 300/700 + 10 × 400/700 = 5.14% + 5.71% = **10.85%**; EVA = 84 − 10.85% × 700 = **₹ 8.05 crore**.

Note (app-written): TYK P15–P16 include the unrecorded patent at market/offer value in invested capital. In TYK P16 ICAI takes cost of debt = 10% (the government bond rate) with no tax adjustment, as no debt rate or tax rate is given (SM p.67).

6.4 Market Value Added (MVA)

- → **MVA = Current Market Value of the firm − Invested Capital.** SM: MV 1,000 − invested capital 920 (Illus 5) = **80**. TYK P14 applies it on equity: market cap − equity funds.
- → Resolves some EVA issues (ignoring **value drivers, book value**). Gives **no basis of share valuation**, but gauges performance efficiency from a **market capitalisation** view — market discounts management's efforts fairly.
- → Higher MVA → "darlings of the share market" → stock becomes 'pricey'; EVA may then have a slightly **negative correlation** with MVA — a **short-term** phenomenon (gap closed by investors), e.g. a stock going **ex-dividend**.

Point (TYK T1)	EVA	MVA
Formula	NOPAT − (Invested Capital × WACC)	Current market value of firm − Invested Capital
Nature of value added	**Derived** value added, for the more discerning investor	**True value added perceived by the market**
Main objective / view	**Management efficiency** in generating returns above the hurdle rate of invested capital	Performance efficiency from a **market capitalisation** point of view; market's view of **future value generation**
Basis	Book-based NOPAT and invested capital (historical)	Market value; attempts to resolve EVA's ignoring of value drivers and book value
Share valuation	Used as a valuation and performance technique	Does **not** itself give a basis of share valuation
Correlation	May be slightly negative in the short term (high-MVA stocks become pricey; ex-dividend); gap closes over time.	

6.5 Shareholder Value Analysis (SVA)

→ Why: NOPAT is **historical**, ignores future earnings, cash flows and **future investment opportunities**. SVA adds **value drivers**: earnings potential (sales), investment opportunities, cost of incremental capital.

→ Steps: (a) FCFs from **value drivers**; (b) discount at **WACC**; (c) add **terminal value**; (d) add current market value of **non-core assets / marketable investments**; (e) deduct **debt** → value of equity.

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SM case ($ mn): EBIT +5%: 100, 105, 110.25, 115.76; interest 5, 6, 7, 8; tax 33% → EAT 63.65, 66.33, 69.18, 72.20; + dep 5, 5, 6, 7 + write-off 1 → OCF **69.65, 71.33, 75.18, 79.20**; − capex (—, 12, 6, 9) − ΔNWC (5, 5, 6, 7) → FCF **64.65, 54.33, 63.18, 63.20**; PVF 12%: 0.89, 0.80, 0.71, 0.64 → PV 57.54, 43.46, 44.86, 40.45; TV = 40.45 × 8.33 (1 ÷ 0.12) = **336.95**; total **482.81** + investment property 35 − debt 19 = equity **498.81**.

7. Arriving at Fair Value

→ Acquirer wants the **fair price — no less no more**. CA: **arm's length**; financial analyst: **PV in cash terms**; speculator: **arbitrage** among similar entities with dissimilar values. In booms (dot com) valuations defy fair value.

→ No single correct method → determine a **range**: **minimum acceptable by seller** to **maximum payable by buyer**; final deal by negotiation (TYK PI range).

→ Exam approach: (i) unless specified, value by **as many methods** as data allow; (ii) **comment** on each; (iii) supplement the conclusion with any additional information.

8. Going Concern and Non-Going Concern Valuation

→ Going concern: no intention or need to liquidate or curtail operations materially; otherwise a different basis, disclosed. Asset valuation depends on it (historical cost).

→ **Non-going concern = Liquidation Valuation**: net value after disposing all assets and discharging all liabilities. **Going concern value = Total Value** (adds value of continuing profits).

→ Going-concern value generally > liquidation value: includes **future profitability, intangibles, goodwill** (premium); liquidation means **laying off employees** and **bad reputation** among investors. Use non-going-concern only when the firm has no value as a going concern.

9. Valuation of Distressed Companies

→ Distress: unable / difficulty paying financial obligations — high fixed costs, illiquid assets, revenues sensitive to downturns (too much debt, or inability to meet operating expenses). Defaulting does not make the company **worthless**.

→ Conventional methods fail: (1) DCF terminal value assumes **infinite life, ever-growing cash flows** — not valid with negative cash flows; (2) **negative, declining revenues**, high bankruptcy risk; DCF values a going concern — projections needed until cash flows turn positive, else negative equity value; (3) **discount rates** reflect sound companies — must be adjusted for probability of failure.

Method	Formula / logic
9.1 Modified DCF	Probability distributions of cash flows (adjust for default); ke from **updated D/E and unlevered beta**; kd from updated default risk. If distribution can't be estimated: **Expected cash flow$_t$ = Cash flow$_t$ × (1 − Probability of distress$_t$)**
9.2 DCF + Distress value	Steps: (i) going-concern value on the path back to health; (ii) probability of distress over DCF life; (iii) distress sale value as % of book value or of DCF value. **Value of Equity = DCF value of equity (1 − Probability of distress) + Distress sale value of equity (Probability of distress)**. Plain DCF **understates** value when the firm may fail and sell assets below PV.
9.3 Adjusted Present Value	Separates investment from financing. **Firm Value = Unlevered Firm Value + (Tax Benefits of Debt − Expected Bankruptcy Cost)**; unlevered value = FCFF at **unlevered cost of equity**; tax benefit = PV of expected tax savings. **Expected Bankruptcy Cost = (Unlevered firm value − Distress sale value) × Probability of distress**
9.4 Relative Valuation	**Revenue and EBITDA multiples** (P/E, P/BV often unusable); adjust industry multiple subjectively: telecom avg 2× revenue → **1.25×** for a distressed telecom firm.

10. Valuation of Start-ups

→ Three common methods: (i) **Earning / cash flow** (discount foreseeable cash flows); (ii) **Asset** (not going concern — liquidation, untimely losses; current realisable values); (iii) **Market** (comparable companies, adjusted). All presuppose an **established**, cash-generating business.

→ Start-up: **no historical data**; value rests on **future growth potential** (often an untested idea) and the founders' competence and drive. Valuation helps investors navigate "using facts, rather than fairy tales".

10.1 Why traditional methods cannot be applied

Method	Why it fails for new-age start-ups
Income	No / minimal **positive cash flows** in the foreseeable future (model accepted by investors).
Asset	(i) Negligible assets — mostly **IP and intangibles**, hard to value and agree; (ii) operate as **going concern**, so value shouldn't be limited to realisable value today.
Market	Disruptors without **established competitors** → numbers skewed, not comparable; some market-approach elements used in **advanced funding rounds**.

10.2 Value Drivers for start-ups

→ **Product** — uniqueness, readiness (functional product/prototype > idea); market testing and customer response.

→ **Management** — credentials and balance of team (engineers + finance + MBA better than only engineers); > half of Indian unicorns have IIT/IIM founders.

→ **Traction** — quantifiable evidence of demand. **Revenue** — more streams, more value. **Industry attractiveness** — logistics, distribution, customer base (tourism start-up less valuable if lockdowns expected).

→ **Demand–supply** — attractive industry → more investor demand. **Competitiveness** — fewer competitors, more value; first-mover advantage (Ola after Uber: easier to convince, but must differentiate).

10.3 Methods for valuing start-ups (10.3.1–10.3.6)

Method	How (SM)
10.3.1 **Berkus** (Dave Berkus)	Five key success factors: (1) basic value, (2) technology, (3) execution, (4) strategic relationships in core market, (5) production and sales. Caps **pre-revenue at $2 mn**, **post-revenue at $2.5 mn**; ignores other market factors; uncomplicated tool.
10.3.2 **Cost-to-Duplicate**	All costs of the start-up and product development incl. physical assets → fair market value. Criticised: ignores **future revenue projections** and assets.
10.3.3 **Comparable Transactions**	"How much were similar start-ups valued at?" XYZ (logistics) acquired ₹ 560 cr with 24 cr users ≈ **₹ 23/user** → ABC with 1.75 cr users ≈ **₹ 40 cr**. Adjust multiplier for proprietary tech, intangibles, penetration, location.
10.3.4 **Scorecard**	Pre-revenue. Average pre-money valuation of comparables × sum of (weight × comparison %). Weights: team 0–30%, opportunity size 0–25%, product/service 0–15%, competitive environment 0–10%, marketing/sales channels/partnerships 0–10%, need for additional investment 0–5%, others 0–5%. On par 100%, below <100%, above >100%; e.g. marketing 10% × 150% = **0.15**.
10.3.5 **First Chicago**	DCF + market approach; value **worst, normal, best case** × probabilities → weighted average value.
10.3.6 **Venture Capital**	Investor seeks a **multiple** of investment (10x, 20x, 30x at exit) or a target IRR for perceived risk; **post-money value** = future value discounted at investor's required return over the relevant time frame.

11. Valuation of Digital Platforms

→ Software-based online infrastructure for **many-to-many** interactions. Types: **Marketplace** (Booking.com, Uber, Amazon), **Search engine** (Google, Bing, Baidu), **Repository** (Spotify, YouTube, GitHub), **Digital communication** (WhatsApp, Teams, Telegram, Slack), **Digital community** (Facebook, LinkedIn), **Payments** (Paytm, GPay).

11.1 Income Approach (11.1.1 Discounting rate, 11.1.2 Specific considerations)

→ **Top-Down:** backward working from **TAM** (Total Addressable Market) → **SAM** (Serviceable Addressable) → **SOM** (Serviceable Obtainable) and time to reach it → business plan. Ambitious for a nascent company. **Bottom-Up:** earnings from limited resources (high-margin sales, cost cutting) — like capital budgeting.

→ Revenue model differs (streaming: subscribers; payments: transactions). Early **cash burn** / low margins to gain share; cash need falls as margins stabilise. Value depends on **quality of forecasts** and promoters' execution.

→ 11.1.1: **FCFF at WACC**, **FCFE at Cost of Equity Capital (Ke)**; Ke by CAPM $R = rf + \beta (rm - rf)$.

→ 11.1.2: (a) beta hard to estimate — unique platforms, scarce listed comparables → use diversified sector, revenue-driving industry or international comparables; (b) survival depends on management quality and adaptability → add a **Company Specific Risk Premium (CSRP) / Alpha** to ke; CSRP **higher for nascent** than mature platforms.

11.2 Market Approach · 11.3 Cost Approach

→ Multiples: P/E, EV/EBITDA, P/BV, Price/Sales; comparables selected by market cap, revenue, margins, capital structure. Difficult because: listed comparables **scarce/absent**; profit and EBITDA may be **negative**; **capital-lite** → very low book value. So emphasis on **revenue drivers**:

Platform	Drivers of revenue
Market place	No. of bookings, registered users, volume of transactions
Payment	Active subscribers, merchants registered, compatibility and speed of OS, security, ease of use
Community	Number of users, subscription fees, platform for professionals
Communication	Number of users, sponsored links, advertising revenue
Repository	Readers and contributors, authenticity, duration of use, quality and variety of data
Search	Number of users, relevant results, time per search (more relevant results in shorter time → premium)

→ Repository: compare users, average views per user, average revenue per user. Example: search platform valued **100.00 Cr** with 50 mn users and 100,00 hours of content used as base for a similar platform with fewer users.

→ **11.3 Cost Approach:** cost to build the same/similar platform — mainly **developer hours spent writing code**; not most appropriate — ignores **revenue-generating capacity**. Fundamentals remain: business understanding, revenue model, management quality, risk-reward.

12. Valuation of Professional / Consultancy Firms

→ Firms giving **customised, knowledge-based services** (CAs, advocates, management consultants); vary widely by service.

→ Compare **historical data** with industry **KPIs and benchmarks** and with competitors; sources: **audited statements, income tax returns**.

→ Income approach: projected growth (TV) matters, but future earnings may not materialise → **factor in the risk**.

→ **Normalisation** of net income and cash flows (add back non-cash and firm-specific items) → compare on equal footing; choose KPIs aligned with acquirer's goals.

13. Impact of ESG on Valuation

→ ESG = framework embedded in strategy to generate value for **all stakeholders**.

→ **E:** climate change, water, waste generation, emissions, biodiversity. **S:** employee development, diversity & inclusion, community development, health & safety, customer. **G:** board independence, board diversity, anti-corruption & bribery, tax transparency, ethical conduct.

→ Developments: ESG funds tapped **> $50 bn** in 2020, ESG assets **> $35 trn**; green bonds crossed **$1 trn** (2020); sustainability **taxonomy** (EU); **IFRS** single global ESG standard; **SEBI** proposed ESG disclosure framework (Feb 2023).

→ High ESG focus → **lower cost of debt**, access to Green / Social / Sustainability-linked bonds; ESG moved from 'good to have' to **'must-to-have'**.

→ Incorporate in DCF via **discount rate** (risk premium — practical but less explicit) or **cash flows** (more explicit): **E** — 2-degree scenario analysis, carbon points; **S** — cost of social measures (labour conditions, CSR, welfare) on revenue; **G** — impact of poor governance (penalty, fines, taxes).

Case Studies

→ **Case 1 — Vodafone / Idea Cellular:** methods: (a) **market value** (NSE price), (b) **comparable companies' market multiple** (BSE/NSE), (c) **NAV** as of 31 Dec 2016; **DCF not used** (no projected cash flows provided) → share exchange. Idea slumped 9.6% (fell up to 14.57%, reversing 14.25% gains) → **'perceived value'** is not quantifiable.

→ **Case 2 — Facebook / WhatsApp:** USD 21.8 bn; 450 mn users at $1/year fee, 70% active, +1 mn users/day (≈ 1 bn in a year); gross per-user value ≈ **USD 55** incl. 4 bn retention sweetener; payback by monetising international messaging.

Exam steps — Relative valuation / EV (Sep 2025 type): $k_e = r_f + \beta (r_m - r_f)$ → adjust EBITDA for extraordinary / pending items → EV = EBITDA × multiple → Equity value = EV − debt + surplus cash (÷ shares) → or EV = shares × CMP + debt − cash.

Exam steps — EVA (Jan 2026 type): PAT ÷ (1 − t) = PBT → + interest = EBIT → NOPAT = EBIT(1 − t) (+ non-cash items) → invested capital (equity + debt, or assets − non-interest CL, + unrecorded patents) → WACC (k_e by CAPM, $k_d(1-t)$, weights) → capital charge → EVA → EVA ÷ shares = max DPS. MVA = market value − invested capital.

Exam steps — DCF / FCFF: high-growth WACC and stable WACC (CAPM + kd(1 − t)) → FCFF = EBIT(1 − t) − (capex − dep) − ΔWC each year → PV at high-growth WACC → TV = terminal-year FCF ÷ (stable cost of capital − growth rate) → discount TV by PVF of the last explicit year → firm value (− debt + non-core assets = equity, as in SVA).

Exam steps — Unlisted firm / new project: proxy β_e → de-gear $\beta_a = \beta_e \times E/[E + D(1 - t)]$ → average → re-gear at own D/E → CAPM ke → WACC → discount FCFs.

Exam steps — Several methods / exchange ratio: NAV per share (deduct contingent liability if it will materialise) → earnings capitalisation (maintainable PAT ÷ rate) → PE × EPS → DCF → weighted / average fair value → range (seller's minimum to buyer's maximum) → exchange ratio = target value ÷ acquirer value; allocate shares on fully-paid equivalents.

Exam steps — Chop-shop / distressed: segment data × pure-play ratios (sales, assets, operating income) → average = break-up value vs market. Distressed: DCF value of equity (1 − Probability of distress) + Distress sale value of equity (Probability of distress); APV = Unlevered Firm Value + Tax Benefits of Debt − (Unlevered firm value − Distress Sale Value) × Probability of Distress.

⑬ Business Valuation — what ICAI asked (Sep 2025, Jan 2026, May 2026)

10 marks *set across the last three attempts (2.8% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.*

Attempt	Question	Marks	Concept tested
Sep 2025	Q1(a)	6	Cost of equity by CAPM, EPS from a profit statement, equity value from an EBITDA multiple, and enterprise value (market cap + debt - cash) *(Numerical)*
Jan 2026	Q4(b)	4	EVA: operating income, EVA with cost of equity from CAPM, and dividend per share that can be paid before value declines *(Numerical)*

What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.

Ⓣ Theory questions (SM Test Your Knowledge) with ICAI answers

Theory Q1

1. Differentiate between EVA and MVA.

ICAI answer: "Please refer paragraph 6.4" – SM paragraph 6.4 reproduced

6.4 Market Value Added (MVA)

The 'MVA' (Market Value Added) simply means the Current Market Value of the firm minus the Invested Capital that we obtained above. Let the current MV of the firm be 1000. Hence MVA will be–

1000 – 920 = 80.

MVA is an attempt to resolve some of the issues involved in EVA e.g., ignoring Value Drivers, Book Value etc. Though MVA itself does not give any basis of share valuation but an alternative way to gauge performance efficiencies of an enterprise, albeit from a market capitalization point of view,

the logic being that the market will discount the efforts taken by the management fairly. Hence, the MVA of 80 arrived in example above is the true value added that is perceived by the market. In contrast, EVA is a derived value added that is for the more discerning investor.

Since MVA represents market views regarding company's future value generation companies with a higher MVA will naturally become the darlings of the share market and would eventually become 'pricey' from a pure pricing perspective. In such cases, the EVA may also sometimes have a slightly negative correlation as compared to MVA. But this will be a short term phenomenon as eventually the gap will get closed by investors themselves. A stock going ex-dividend will exhibit such propensities.

We can conclude that the main objective of EVA is thus to show management efficiency in generating returns over and above the hurdle rate of invested capital.

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Theory Q2

2. Relative Valuation is the method to arrive at a 'relative' value using a 'comparative' analysis to its peers or similar enterprises. Elaborate this statement.

ICAI answer: "Please refer paragraph 5" – SM paragraph 5 reproduced

5. RELATIVE VALUATION

The three approaches that we saw to arriving at the value of an enterprise viz. the asset based, the earnings based and the cash flow based are for arriving at the 'intrinsic value' of the same. Relative Valuation is the method to arrive at a 'relative' value using a 'comparative' analysis to its peers or similar enterprises. However, increasingly the contemporary financial analysts are using relative valuation in conjunction to the afore-stated approaches to validate the intrinsic value arrived earlier.

The Concept of 'Relative Valuation': One way to look at the practical implementation of fair value within the valuation context would be to identify assets that are similar to the ones held by the acquiree company so that the values can be compared. This would be a significant departure from the 'intrinsic value' approach that we have seen until now. Trying to get a value that would be the nearest to the market price would mean that the valuation of a particular portfolio, or a divestiture in an entity, would happen at an agreeable price that fits into the normal distribution.

In one sense, we are indeed using the relative valuation in a limited approach when we speak about expected market returns, or when we are adopting an index based comparative. The more the asset pricing gets correlated to the similar assets in the market, the more inclusive it gets. Thus, when we are comparing bonds, the closer the YTM of the bond to the government index of return, the more credible it gets when it comes to pricing.

The Relative valuation, also referred to as 'Valuation by multiples,' uses financial ratios to derive at the desired metric (referred to as the 'multiple') and then compares the same to that of comparable firms. Comparable firms would mean the ones having similar asset and risk dispositions and assumed to continue to do so over the comparison period. In the process, there may be extrapolations set to the desired range to achieve the target set. To elaborate –

1. Find out the 'drivers' that will be the best representative for deriving at the multiple
2. Determine the results based on the chosen driver(s) through financial ratios
3. Find out the comparable firms, and perform the comparative analysis, and,

4. Iterate the value of the firm obtained to smoothen out the deviations

Step 1: Finding the correct driver that goes to determine the multiple is significant for relative valuation as it sets the direction to the valuation approach. Thereby, one can have two sets of multiple based approaches depending on the types of the drivers –

(a) Enterprise value based multiples, which would consist primarily of EV/EBITDA, EV/Invested Capital and EV/Sales.

(b) Equity value based multiples, which would comprise of P/E ratio and Price Earning Growth (PEG) Ratio.

We have already seen the concept and application of Enterprise Value in previous section. However, in light of relative valuation, we can definitely add that whereas EV/EBITDA is a popular ratio and does provide critical inputs, the EV/Invested Capital will be more appropriate to capital intensive enterprises, and EV/Sales will be used by companies who are cash rich, have a huge order book, and forecast organic growth through own capital.

The P/E has a celebrated status amongst Equity based multiples, and the PEG (PE Ratio/ Growth Rate i.e. the ratio of the PE to the expected growth rate of the firm) is more suitable where we are doing relative valuation of either high growth or sunrise industries.

Step 2: Choosing the right financial ratio is a vital part of success of this model. A factor based approach may help in getting this correct – for example – a firm that generates revenue mostly by exports will be highly influenced by future foreign exchange fluctuations. A pure P/E based ratio may not be reflective of this reality, which couldn't pre-empt the impacts that Brexit triggered on currency values. Likewise, an EV/Invested Capital would be a misfit for a company which may be light on core assets, or if has significant investment properties.

Step 3: Arriving at the right mix of comparable firms. This is perhaps the most challenging of all the steps – No two entities can be same – even if they may seem to be operating within the same risk and opportunity perimeter. So, a software company 'X' that we are now comparing to a similar sized company 'Y' may have a similar capital structure, a similar operative environment, and head count size – so far the two firms are on even platform for returns forecast and beta values. On careful scrutiny, it may be realized that the revenue generators are different – X may be deriving its revenues from dedicated service contracts having Full Time Equivalent (FTE) pricing, whereas Y earns through Unit Transfer Pricing (UTP) model. This additional set of information dramatically changes the risk structure – and this is precisely what the discerning investor has to watch for. In other words, take benchmarks with a pinch of salt.

Take another example – a firm is operating in a niche market, and that obviously leads to getting comparable firms become a difficult task. In such cases, one may have to look beyond the current operating market and identify similar structured companies from other industries.

The comparable firm can either be from a peer group operating within the same risks and opportunities perimeter, or alternatively can be just take closely relevant firms and then perform a regression to arrive at the comparable metrics. You would notice that in our example, the analyst is adopting the later approach. Whereas the company 'X' will have to ignore 'Y' and search for a similar revenue-risk based company. However, as a last resort, it may adopt a regression based model as above.

Step 4: Iterate / extrapolate the results obtained to arrive at the correct estimate of the value of the firm.

Thus, we can conclude that 'Relative Valuation' is a comparative driven approach that assumes that the value of similar firms can form a good indicator for the value of the tested firm. There are some assumptions that are inherent to this model –

i. The market is efficient

ii. The function between the fundamentals and the multiples are linear

iii. The firms that are comparable are similar in structure, risk and growth pattern

Further, we can approach Enterprise Value (EV) in two ways –

(a) Take Entity Value as the base, and then adjust for debt values for arriving the 'EV';

or

(b) Take a balance sheet based approach and arrive at EV.

Let's apply the above concepts into a relative valuation illustration:

Illustration 2

A Ltd. made a Gross Profit of ₹ 10,00,000 and incurred Indirect Expenses of ₹ 4,00,000. The number of issued Equity Shares is 1,00,000. The company has a Debt of ₹ 3,00,000 and Surplus Funds to the tune of ₹ 5,00,000. The market related details are as follows:

Risk Free Rate of Return	4.5%
Market Rate of Return	12%
β of the Company	0.9

Determine:

(a) Per Share Earning Value of the Company.

(b) Equity Value of the company if applicable EBITDA multiple is 5.

Solution

(a) Capitalization Rate using CAPM

4.5% + 0.9(12% - 4.5%) = 11.25%

Calculation of Earning Value Per Share

	(₹ 000)
Gross Profit	1000
Less: Indirect Expenses	(400)
EBITDA	600
Earning Value of Company (600/ 0.1125)	5333.33
Number of Shares	1,00,000
Earning Value Per Share	₹ 53.33

(b) Equity Value of Company

	(₹ 000)
EBITDA	600
EBITDA Multiple	5
Capitalized Value	3000
Less: Debt	(300)
Add: Surplus Funds	500
Equity Value	3200

Now let us see how EV can be arrived at using Balance Sheet approach in the following illustration.

Illustration 3

The balance sheet of H K Ltd. is as follows:

	₹ 000
Non-Current Assets	1000
Current Assets	
Trade Receivables	500

Cash and cash equivalents	500
	2000
Shareholders' funds	800
Long Term Debt	200
Current Liabilities and Provisions	1000
	2000

The shares are actively traded and the Current Market Price (CMP) is ₹ 12 per share. Shareholder funds represent 70,000 shares of ₹ 10 each and rest is retained earnings. Calculate the Enterprise Value of HK Ltd.

Solution

Shares outstanding	70,000
CMP	₹ 12
Market Capitalization	₹ 8,40,000
Add: Debt	₹ 2,00,000
Less: Cash & Cash equivalents	(₹ 5,00,000)
Enterprise Value (EV)	**₹ 5,40,000**

ICAI answer: "Please refer paragraph 5" – SM paragraph 5 reproduced

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Take another example – a firm is operating in a niche market, and that obviously leads to getting comparable firms become a difficult task. In such cases, one may have to look beyond the current operating market and identify similar structured companies from other industries.

The comparable firm can either be from a peer group operating within the same risks and opportunities perimeter, or alternatively can be just take closely relevant firms and then perform a regression to arrive at the comparable metrics. You would notice that in our example, the analyst is adopting the later approach. Whereas the company 'X' will have to ignore 'Y' and search for a similar revenue-risk based company. However, as a last resort, it may adopt a regression based model as above.

Step 4: Iterate / extrapolate the results obtained to arrive at the correct estimate of the value of the firm.

Thus, we can conclude that 'Relative Valuation' is a comparative driven approach that assumes that the value of similar firms can form a good indicator for the value of the tested firm. There are some assumptions that are inherent to this model –

i. The market is efficient

ii. The function between the fundamentals and the multiples are linear

iii. The firms that are comparable are similar in structure, risk and growth pattern

Further, we can approach Enterprise Value (EV) in two ways –

(a) Take Entity Value as the base, and then adjust for debt values for arriving the 'EV';

or

(b) Take a balance sheet based approach and arrive at EV.

Let's apply the above concepts into a relative valuation illustration:

Illustration 2

A Ltd. made a Gross Profit of ₹ 10,00,000 and incurred Indirect Expenses of ₹ 4,00,000. The number of issued Equity Shares is 1,00,000. The company has a Debt of ₹ 3,00,000 and Surplus Funds to the tune of ₹ 5,00,000. The market related details are as follows:

Risk Free Rate of Return	4.5%
Market Rate of Return	12%
β of the Company	0.9

Determine:

(a) Per Share Earning Value of the Company.

(b) Equity Value of the company if applicable EBITDA multiple is 5.

Solution

(a) Capitalization Rate using CAPM

4.5% + 0.9(12% - 4.5%) = 11.25%

Calculation of Earning Value Per Share

	(₹ 000)
Gross Profit	1000
Less: Indirect Expenses	(400)
EBITDA	600
Earning Value of Company (600/ 0.1125)	5333.33
Number of Shares	1,00,000
Earning Value Per Share	₹ 53.33

(b) Equity Value of Company

	(₹ 000)
EBITDA	600
EBITDA Multiple	5
Capitalized Value	3000
Less: Debt	(300)
Add: Surplus Funds	500
Equity Value	3200

Now let us see how EV can be arrived at using Balance Sheet approach in the following illustration.

Illustration 3

The balance sheet of H K Ltd. is as follows:

	₹ 000
Non-Current Assets	1000
Current Assets	
Trade Receivables	500

Cash and cash equivalents	500
	2000
Shareholders' funds	800
Long Term Debt	200
Current Liabilities and Provisions	1000
	2000

The shares are actively traded and the Current Market Price (CMP) is ₹ 12 per share. Shareholder funds represent 70,000 shares of ₹ 10 each and rest is retained earnings. Calculate the Enterprise Value of HK Ltd.

Solution

Shares outstanding	70,000
CMP	₹ 12
Market Capitalization	₹ 8,40,000
Add: Debt	₹ 2,00,000
Less: Cash & Cash equivalents	(₹ 5,00,000)
Enterprise Value (EV)	**₹ 5,40,000**

① SM illustrations with ICAI solutions

4. Measuring Cost Of Equity

SM Ch 13 · Illustration 1 (under 4.3 Estimating Beta and Valuation of Unlisted Companies)

Illustration 1

There is a privately held company X Pvt. Ltd that is operating into the retail space, and is now scouting for angel investors. The details pertinent to valuing X Pvt. Ltd are as follows –

The company has achieved break even this year and has an EBITDA of 90. The unleveraged beta based on the industry in which it operates is 1.8, and the average debt to equity ratio is hovering at 40:60. The rate of return provided by risk free liquid bonds is 5%. The EV is to be taken at a multiple of 5 on EBITDA. The accountant has informed that the EBITDA of 90 includes an extraordinary gain of 10 for the year, and a potential write off of preliminary sales promotion costs of 20 are still pending. The internal assessment of rate of market return for the industry is 11%. The FCFs for the next 3 years are as follows:

	Y1	Y2	Y3
Future Cash flows	100	120	150

The pre-tax cost of debt is 12%. Assume a tax regime of 30%.

What is the potential value to be placed on X Pvt. Ltd?

Solution — ICAI

Solution

The levered beta of the company will be 1.8[1+(1-0.3)*40/60)] = 2.64

The adjusted EBITDA would be 90 –10 – 20 = 60

The EV will be multiple of 5 on the 60 obtained above = 300

The Cost of equity in accordance with CAPM = r (f) + β (Rm – Rf)

= 5% + 2.64 (11% - 5%) = 20.84%

The WACC = Cost of Equity + Cost of Debt

= 20.84 (60/100) + 12.0 (1-0.3) (40/100) = 15.864

Finally, the future cash flows can be discounted at the WACC obtained above as under –

	Y1	Y2	Y3
Future Cash flows	100	120	150
Discount factor	0.863	0.745	0.643
PVs of cash flows	86.30	89.40	96.45
VALUE OF THE FIRM			**272.15**

5. Relative Valuation

SM Ch 13 · Illustration 2

Illustration 2

A Ltd. made a Gross Profit of ₹ 10,00,000 and incurred Indirect Expenses of ₹ 4,00,000. The number of issued Equity Shares is 1,00,000. The company has a Debt of ₹ 3,00,000 and Surplus Funds to the tune of ₹ 5,00,000. The market related details are as follows:

Risk Free Rate of Return	4.5%
Market Rate of Return	12%
β of the Company	0.9

Determine:

(a) Per Share Earning Value of the Company.

(b) Equity Value of the company if applicable EBITDA multiple is 5.

Solution — ICAI

Solution

(a) Capitalization Rate using CAPM

4.5% + 0.9(12% - 4.5%) = 11.25%

Calculation of Earning Value Per Share

	(₹ 000)
Gross Profit	1000
Less: Indirect Expenses	(400)
EBITDA	600
Earning Value of Company (600/ 0.1125)	5333.33
Number of Shares	1,00,000
Earning Value Per Share	₹ 53.33

(b) Equity Value of Company

	(₹ 000)
EBITDA	600
EBITDA Multiple	5
Capitalized Value	3000
Less: Debt	(300)
Add: Surplus Funds	500
Equity Value	3200

Now let us see how EV can be arrived at using Balance Sheet approach in the following illustration.

SM Ch 13 · Illustration 3

Illustration 3

The balance sheet of H K Ltd. is as follows:

	₹ 000
Non-Current Assets	1000
Current Assets	
Trade Receivables	500
Cash and cash equivalents	500
	2000
Shareholders' funds	800
Long Term Debt	200
Current Liabilities and Provisions	1000
	2000

The shares are actively traded and the Current Market Price (CMP) is ₹ 12 per share. Shareholder funds represent 70,000 shares of ₹ 10 each and rest is retained earnings. Calculate the Enterprise Value of HK Ltd.

Solution — ICAI

Solution

Shares outstanding	70,000
CMP	₹ 12
Market Capitalization	₹ 8,40,000
Add: Debt	₹ 2,00,000
Less: Cash & Cash equivalents	(₹ 5,00,000)
Enterprise Value (EV)	**₹ 5,40,000**

6. Other Approaches To Value Measurement

SM Ch 13 · Illustration 4 (under 6.2 Chop-Shop Method)

Illustration 4

Using the chop-shop approach (or Break-up value approach), assign a value for Cornett GMBH. whose stock is currently trading at a total market price of €4 million. For Cornett, the accounting data set forth in three business segments: consumer wholesaling, specialty services, and assorted centers. Data for the firm's three segments are as follows:

Business segment	Segment sales	Segment assets	Segment income
Consumer wholesaling	€1,500,000	€ 750,000	€100,000
Specialty services	€800,000	€700,000	€150,000
Assorted centers	€2,000,000	€3,000,000	€600,000

Industry data for "pure-play" firms have been compiled and are summarized as follows:

Business segment	Capitalization/sales	Capitalization/assets	Capitalization/ operating income
Consumer wholesaling	0.75	0.60	10.00
Specialty services	1.10	0.90	7.00
Assorted centers	1.00	0.60	6.00

Solution — ICAI

Solution

Cornett, GMBH. – Break-up valuation

Business Segment	Capital-to-Sales	Segment Sales	Theoretical Values
Consumer wholesaling	0.75	€1,500,000	€1,125,000
Specialty services	1.10	€800,000	€880,000
Assorted centers	1.00	€2,000,000	€2,000,000
Total value			€4,005,000

Business Segment	Capital-to-Sales	Segment Sales	Theoretical Values
Consumer wholesaling	0.60	€750,000	€450,000
Specialty services	0.90	€700,000	€630,000
Assorted centers	0.60	€3,000,000	€1,800,000
Total value			€2,880,000

Business Segment	Capital-to-Sales	Segment Sales	Theoretical Values
Consumer wholesaling	10.00	€100,000	€1,000,000
Specialty services	7.00	€150,000	€1,050,000
Assorted centers	6.00	€600,000	€3,600,000
Total value			€5,650,000

$$\text{Average theoretical value} = \frac{4,005,000 + 2,880,000 + 5,650,000}{3} = 4,178,333.33 \text{ say } 4,178,000$$

Average theoretical value of Cornett GMBH. = €4,178,000

SM Ch 13 · Illustration 5 (under 6.3 Economic Value Added (EVA))

Illustration 5

Compute EVA of A Ltd. with the following information:

All Figure are in ₹ Lac

Profit and Loss Statement		Balance Sheet	
Revenue	1000	PPE	1000
Direct Costs	-390	Current Assets	300
Selling, General & Admin. Exp. (SGA)	-200		**1300**
EBIT	**410**	Equity	700
Interest	-10	Reserves	100
EBT	**400**	Non-Current Borrowings	100
Tax Expense	-120	Current Liabilities & Provisions	400
EAT	**280**		**1300**

Assume Bad Debts provision of ₹ 20 Lac is included in the SGA, and same amount is reduced from the trade receivables in current assets.

Also assume that the pre-tax Cost of Debt is 12%, Tax Rate is 30% and Cost of Equity (i.e. shareholder's expected return) is 8.45%.

Solution — ICAI

Solution

Step I: Computation of NOPAT

NOPAT	
EBIT	410
Less: Taxes	-123
Add: Non-Cash Expenses	20
NOPAT	**307**

Step II: Finding out the Invested Capital:

Invested Capital	
Total Assets	1300
Less: Non Interest bearing liabilities	-400
	900
Add: Non Cash adjustment	20
	920

Note: It is assumed that the current liabilities also include the 100 of tax liability.

Step III: Compute the WACC

WACC = Cost of equity + Cost of debt

In this case, WACC = (800/900*8.45%) + [100/900*12% (1 - 0.30)] = 8.44%

Step IV: Find out the Capital Charge

Capital Charge = Invested Capital * WACC = 920 * 8.44% = 77.65

Step V: EVA = Adjusted NOPAT – Capital Charge = 307 – 77.65 = 229.35

Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

Practical QI

1. ABC Company is considering acquisition of XYZ Ltd. which has 1.5 crores shares outstanding and issued. The market price per share is ₹ 400 at present. ABC's average cost of capital is 12%. Available information from XYZ indicates its expected cash accruals for the next 3 years as follows:

Year	₹ Cr.
1	250
2	300
3	400

Calculate the range of valuation that ABC has to consider. (PV factors at 12% for years 1 to 3 respectively: 0.893, 0.797 and 0.712).

Solution — ICAI

1. VALUATION BASED ON MARKET PRICE

Market Price per share ₹ 400

Thus value of total business is (₹ 400 x 1.5 Cr.) ₹ 600 Cr.

VALUATION BASED ON DISCOUNTED CASH FLOW

Present Value of cash flows

(₹ 250 cr x 0.893) + (₹ 300 cr. X 0.797) + (₹ 400 cr. X 0.712) = ₹ 747.15 Cr.

Value of per share (₹ 747.15 Cr. / 1.5 Cr) ₹ 498.10 per share

RANGE OF VALUATION

	Per Share ₹	Total ₹ Cr.
Minimum	400.00	600.00
Maximum	498.10	747.15

Practical Q2

2. Eagle Ltd. reported a profit of ₹ 77 lakhs after 30% tax for the financial year 2011-12. An analysis of the accounts revealed that the income included extraordinary items of ₹ 8 lakhs and an extraordinary loss of ₹10 lakhs. The existing operations, except for the extraordinary items, are expected to continue in the future. In addition, the results of the launch of a new product are expected to be as follows:

	₹ In lakhs
Sales	70
Material costs	20
Labour costs	12
Fixed costs	10

You are required to:

(i) Calculate the value of the business, given that the capitalization rate is 14%.

(ii) Determine the market price per equity share, with Eagle Ltd.'s share capital being comprised of 1,00,000 13% preference shares of ₹ 100 each and 50,00,000 equity shares of ₹ 10 each and the P/E ratio being 10 times.

Solution — ICAI

2. (i) Computation of Business Value

			(₹ Lakhs)
Profit before tax $\frac{77}{1-0.30}$			110
Less: Extraordinary income			(8)
Add: Extraordinary losses			10
			112
Profit from new product		(₹ Lakhs)	
Sales		70	
Less: Material costs	20		
Labour costs	12		
Fixed costs	10	(42)	28
			140.00
Less: Taxes @30%			42.00
Future Maintainable Profit after taxes			98.00
Relevant Capitalisation Factor			0.14
Value of Business (₹98/0.14)			700

(ii) Determination of Market Price of Equity Share

Future maintainable profits (After Tax)	₹ 98,00,000
Less: Preference share dividends 1,00,000 shares of ₹ 100 @ 13%	₹ 13,00,000
Earnings available for Equity Shareholders	₹ 85,00,000

No. of Equity Shares	50,00,000
Earning per share = $\frac{₹\ 85,00,000}{50,00,000}$ =	₹ 1.70
PE ratio	10
Market price per share	₹ 17

Practical Q3

3. ABC Co. is considering a new sales strategy that will be valid for the next 4 years. They want to know the value of the new strategy. Following information relating to the year which has just ended, is available:

Income Statement	**₹**
Sales	20,000
Gross margin (20%)	4,000
Administration, Selling & distribution expense (10%)	2,000
PBT	2,000
Tax (30%)	600
PAT	1,400
Balance Sheet Information	
Fixed Assets	8,000
Current Assets	4,000
Equity	12,000

If it adopts the new strategy, sales will grow at the rate of 20% per year for three years. From 4th year onward Cash Flow will be stabilized. The gross margin ratio, Assets turnover ratio, the Capital structure and the income tax rate will remain unchanged.

Depreciation would be at 10% of net fixed assets at the beginning of the year.

The Company's target rate of return is 15%.

Determine the incremental value due to adoption of the strategy.

Solution — ICAI

3. Projected Balance Sheet

	Year 1	Year 2	Year 3	Year 4
Fixed Assets (40% of Sales)	9,600	11,520	13,824	13,824
Current Assets (20% of Sales)	4,800	5,760	6,912	6,912
Total Assets	14,400	17,280	20,736	20,736
Equity	14,400	17,280	20,736	20,736

Projected Cash Flows:-

	Year 1	Year 2	Year 3	Year 4
Sales	24,000	28,800	34,560	34,560
PBT (10% of sale)	2,400	2,880	3,456	3,456
PAT (70%)	1,680	2,016	2,419.20	2,419.20
Depreciation	800	960	1,152	1,382
Addition to Fixed Assets	2,400	2,880	3,456	1,382
Increase in Current Assets	800	960	1,152	-
Operating cash flow (FCFF)	(720)	(864)	(1,036.80)	2,419.20

Projected Cash Flows:-

Present value of Projected Cash Flows:-

Cash Flows	*PVF at 15%*	*PV*
-720	0.870	-626.40
-864	0.756	-653.18
-1,036.80	0.658	-682.21
		-1,961.79

Residual Value - 2419.20/0.15 = 16,128

Present value of Residual value	=	$16128/(1.15)^3$
	=	16128/1.521 = 10603.55
Total shareholders' value	=	10,603.55 – 1,961.79 = 8,641.76
Pre strategy value	=	1,400 / 0.15 = 9,333.33
∴ Value of strategy	=	8,641.76 – 9,333.33 = – 691.57

Conclusion: The strategy is not financially viable

Practical Q4

4. H Ltd. agrees to buy over the business of B Ltd. effective 1st April, 2012.The summarized Balance Sheets of H Ltd. and B Ltd. as on 31st March 2012 are as follows:

Balance sheet as at 31st March, 2012 (In Crores of Rupees)

Liabilities:	H. Ltd	B. Ltd.
Paid up Share Capital		
-Equity Shares of ₹100 each	350.00	--
-Equity Shares of ₹10 each	--	6.50
Reserve & Surplus	950.00	25.00
Total	**1,300.00**	**31.50**
Assets:		
Net Fixed Assets	220.00	0.50
Net Current Assets	1,020.00	29.00
Deferred Tax Assets	60.00	2.00
Total	**1,300.00**	**31.50**

H Ltd. proposes to buy out B Ltd. and the following information is provided to you as part of the scheme of buying:

(1) The weighted average post tax maintainable profits of H Ltd. and B Ltd. for the last 4 years are ₹ 300 crores and ₹ 10 crores respectively.

(2) Both the companies envisage a capitalization rate of 8%.

(3) H Ltd. has a contingent liability of ₹ 300 crores as on 31st March, 2012.

(4) H Ltd. to issue shares of ₹ 100 each to the shareholders of B Ltd. in terms of the exchange ratio as arrived on a Fair Value basis. (Please consider weights of 1 and 3 for the value of shares arrived on Net Asset basis and Earnings capitalization method respectively for both H Ltd. and B Ltd.)

You are required to arrive at the value of the shares of both H Ltd. and B Ltd. under:

(i) Net Asset Value Method

(ii) Earnings Capitalisation Method

(iii) Exchange ratio of shares of H Ltd. to be issued to the shareholders of B Ltd. on a Fair value basis (taking into consideration the assumption mentioned in point 4 above.)

Solution — ICAI

4. (i) Net asset value

H Ltd.	$\frac{₹\ 1300\ \text{Crores} - ₹\ 300\ \text{Crores}}{3.50\ \text{Crores}} = ₹\ 285.71$
B Ltd.	$\frac{₹\ 31.50\ \text{Crores}}{0.65\ \text{Crores}} = ₹\ 48.46$

(ii) Earning capitalization value

H Ltd.	$\frac{₹\ 300\ \text{Crores} / 0.08}{3.50\ \text{Crores}} = ₹\ 1071.43$
B Ltd.	$\frac{₹\ 10\ \text{Crores} / 0.08}{0.65\ \text{Crores}} = ₹\ 192.31$

(iii) Fair value

H Ltd.	$\frac{₹\ 285.71 \times 1 + ₹\ 1071.43 \times 3}{4} = ₹\ 875$
B Ltd.	$\frac{₹\ 48.46 \times 1 + ₹\ 192.31 \times 3}{4} = ₹\ 156.3475$
Exchange ratio	₹156.3475/ ₹875 = 0.1787

H Ltd should issue its 0.1787 share for each share of B Ltd.

Note: In above solution it has been assumed that the contingent liability will materialize at its full amount.

Practical QS

5. AB Ltd., is planning to acquire and absorb the running business of XY Ltd. The valuation is to be based on the recommendation of merchant bankers and the consideration is to be discharged in the form of equity shares to be issued by AB Ltd. As on 31.3.2006, the paid up capital of AB Ltd. consists of 80 lakhs shares of ₹ 10 each. The highest and the lowest market quotation during the last 6 months were ₹ 570 and ₹ 430. For the purpose of the exchange, the price per share is to be reckoned as the average of the highest and lowest market price during the last 6 months ended on 31.3.06.

XY Ltd.'s Balance Sheet as at 31.3.2006 is summarised below:

	₹ lakhs
Sources	
Share Capital	
20 lakhs equity shares of ₹10 each fully paid	200
10 lakhs equity shares of ₹10 each, ₹5 paid	50
Loans	100
Total	350
Uses	
Fixed Assets (Net)	150
Net Current Assets	200
	350

An independent firm of merchant bankers engaged for the negotiation, have produced the following estimates of cash flows from the business of XY Ltd.:

Year ended	By way of	₹ lakhs
31.3.07	after tax earnings for equity	105
31.3.08	do	120
31.3.09	Do	125
31.3.10	Do	120
31.3.11	Do	100
	Terminal Value estimate	200

It is the recommendation of the merchant banker that the business of XY Ltd. may be valued on the basis of the average of (i) Aggregate of discounted cash flows at 8% and (ii) Net assets value. Present value factors at 8% for years

1-5: 0.93 0.86 0.79 0.74 0.68

You are required to:

(i) Calculate the total value of the business of XY Ltd.

(ii) The number of shares to be issued by AB Ltd.; and

(iii) The basis of allocation of the shares among the shareholders of XY Ltd.

Solution — ICAI

5. Price/share of AB Ltd. for determination of number of shares to be issued

= (₹ 570 + ₹ 430)/2 = ₹ 500

Value of XY Ltd based on future cash flow capitalization (105x0.93)+(120x0.86)+(125x0.79)+(120x0.74)x(300x0.68)	₹ lakhs	592.40
Value of XY Ltd based on net assets	₹ lakhs	250.00
Average value (592.40+250)/2		421.20

No. of shares in AB Ltd to be issued ₹ 4,21,20,000/500	Nos.	84240
Basis of allocation of shares		
Fully paid equivalent shares in XY Ltd. (20+5) lakhs		2500000
Distribution to fully paid shareholders 84240x20/25		67392
Distribution to partly paid shareholders 84240-67392		16848

Practical Q6

6. The valuation of Hansel Limited has been done by an investment analyst. Based on an expected free cash flow of ₹ 54 lakhs for the following year and an expected growth rate of 9 percent, the analyst has estimated the value of Hansel Limited to be ₹ 1800 lakhs. However, he committed a mistake of using the book values of debt and equity.

The book value weights employed by the analyst are not known, but you know that Hansel Limited has a cost of equity of 20 percent and post tax cost of debt of 10 percent. The value of equity is thrice its book value, whereas the market value of its debt is nine-tenths of its book value. What is the correct value of Hansel Ltd?

Solution — ICAI

6. Cost of capital by applying Free Cash Flow to Firm (FCFF) Model is as follows:-

$$\text{Value of Firm} = V_0 = \frac{FCFF_1}{K_c - g_n}$$

Where –

$FCFF_1$ = Expected FCFF in the year 1

K_c= Cost of capital

g_n = Growth rate forever

Thus, ₹ 1800 lakhs = ₹ 54 lakhs /(K_c-g)

Since g = 9%, then K_c = 12%

Now, let X be the weight of debt and given cost of equity = 20% and cost of debt = 10%, then 20% (1 – X) + 10% X = 12%

Hence, X = 0.80, so book value weight for debt was 80%

∴ Correct weight should be 60 of equity and 72 of debt.

∴ Cost of capital = K_c = 20% (60/132) + 10% (72/132) = 14.5455% and correct firm's value

= ₹ 54 lakhs/(0.1454 – 0.09) = ₹ 974.73 lakhs.

Practical Q7

7. Following information are available in respect of XYZ Ltd. which is expected to grow at a higher rate for 4 years after which growth rate will stabilize at a lower level:

Base year information:

Revenue	- ₹ 2,000 crores
EBIT	- ₹ 300 crores
Capital expenditure	- ₹ 280 crores
Depreciation	- ₹ 200 crores

Information for high growth and stable growth period are as follows:

	High Growth	Stable Growth
Growth in Revenue & EBIT	20%	10%
Growth in capital expenditure and depreciation	20%	Capital expenditure are offset by depreciation
Risk free rate	10%	9%
Equity beta	1.15	1
Market risk premium	6%	5%
Pre tax cost of debt	13%	12.86%
Debt equity ratio	1 : 1	2 : 3

For all time, working capital is 25% of revenue and corporate tax rate is 30%.

What is the value of the firm?

Solution — ICAI

7. **High growth phase :**

k_e = 0.10 + 1.15 x 0.06 = 0.169 or 16.9%.

k_d = 0.13 x (1-0.3) = 0.091 or 9.1%.

Cost of capital = 0.5 x 0.169 + 0.5 x 0.091 = 0.13 or 13%.

Stable growth phase :

k_e = 0.09 + 1.0 x 0.05 = 0.14 or 14%.

k_d = 0.1286 x (1 - 0.3) = 0.09 or 9%.

Cost of capital = 0.6 x 0.14 + 0.4 x 0.09 = 0.12 or 12%.

Determination of forecasted Free Cash Flow of the Firm (FCFF)

(₹ in crores)

	Yr. 1	Yr. 2	Yr. 3	Yr. 4	Terminal Year
Revenue	2,400	2,880	3,456	4,147.20	4,561.92
EBIT	360	432	518.40	622.08	684.29
EAT	252	302.40	362.88	435.46	479.00
Capital Expenditure	96	115.20	138.24	165.89	-
Less Depreciation					
Δ Working Capital	100.00	120.00	144.00	172.80	103.68
Free Cash Flow (FCF)	56.00	67.20	80.64	96.77	375.32

Alternatively, it can also be computed as follows:

(₹ in crores)

	Yr. 1	Yr. 2	Yr. 3	Yr. 4	Terminal Year
Revenue	2,400	2,880	3,456	4,147.20	4,561.92
EBIT	360	432	518.40	622.08	684.29
EAT	252	302.40	362.88	435.46	479.00
Add: Depreciation	240	288	345.60	414.72	456.19
	492	590.40	708.48	850.18	935.19
Less: Capital Exp.	336	403.20	483.84	580.61	456.19
Δ WC	100.00	120.00	144.00	172.80	103.68
	56.00	67.20	80.64	96.77	375.32

Present Value (PV) of FCFF during the explicit forecast period is:

FCFF (₹ in crores)	PVF @ 13%	PV (₹ in crores)
56.00	0.885	49.56
67.20	0.783	52.62
80.64	0.693	55.88
96.77	0.613	59.32
		₹ 217.38

Terminal Value of Cash Flow

$$\frac{375.32}{0.12-0.10} = ₹\,18{,}766.00\,\text{Crores}$$

PV of the terminal, value is:

$$₹\,18{,}766.00\,\text{Crores} \times \frac{1}{(1.13)^4} = ₹\,18{,}766.00\,\text{Crores} \times 0.613 = ₹\,11{,}503.56\,\text{Crores}$$

The value of the firm is :

₹ 217.38 Crores + ₹ 11,503.56 Crores = ₹ 11,720.94 Crores

Practical Q8

8. Following information is given in respect of WXY Ltd., which is expected to grow at a rate of 20% p.a. for the next three years, after which the growth rate will stabilize at 8% p.a. normal level, in perpetuity.

	For the year ended March 31, 2014
Revenues	₹ 7,500 Crores
Cost of Goods Sold (COGS)	₹ 3,000 Crores
Operating Expenses	₹ 2,250 Crores
Capital Expenditure	₹ 750 Crores
Depreciation (included in Operating Expenses)	₹ 600 Crores

During high growth period, revenues & Earnings before Interest & Tax (EBIT) will grow at 20% p.a. and capital expenditure net of depreciation will grow at 15% p.a. From year 4 onwards, i.e. normal growth period revenues and EBIT will grow at 8% p.a. and incremental capital expenditure will be offset by the depreciation. During both high growth & normal growth period, net working capital requirement will be 25% of revenues.

The Weighted Average Cost of Capital (WACC) of WXY Ltd. is 15%.

Corporate Income Tax rate will be 30%.

Required:

Estimate the value of WXY Ltd. using Free Cash Flows to Firm (FCFF) & WACC methodology.

The PVIF @ 15 % for the three years are as below:

Year	t_1	t_2	t_3
PVIF	0.8696	0.7561	0.6575

Solution — ICAI

8. Determination of forecasted Free Cash Flow of the Firm (FCFF)

(₹ in crores)

	Yr. 1	Yr. 2	Yr. 3	Terminal Year
Revenue	9000.00	10800.00	12960.00	13996.80
COGS	3600.00	4320.00	5184.00	5598.72
Operating Expenses	1980.00*	2376.00	2851.20	3079.30
Depreciation	720.00	864.00	1036.80	1119.74
EBIT	2700.00	3240.00	3888.00	4199.04
Tax @30%	810.00	972.00	1166.40	1259.71
EAT	1890.00	2268.00	2721.60	2939.33
Capital Exp. – Dep.	172.50	198.38	228.13	-
Δ Working Capital	375.00	450.00	540.00	259.20
Free Cash Flow (FCF)	1342.50	1619.62	1953.47	2680.13

* Excluding Depreciation.

Present Value (PV) of FCFF during the explicit forecast period is:

FCFF (₹ in crores)	PVF @ 15%	PV (₹ in crores)
1342.50	0.8696	1167.44
1619.62	0.7561	1224.59
1953.47	0.6575	1284.41
		3676.44

PV of the terminal, value is:

$$\frac{2680.13}{0.15 - 0.08} \times \frac{1}{(1.15)^3} = ₹\ 38287.57 \text{ Crore} \times 0.6575 = ₹\ 25174.08 \text{ Crore}$$

The value of the firm is :

₹ 3676.44 Crores + ₹ 25174.08 Crores = ₹ 28,850.52 Crores

Practical Q9

9. With the help of the following information of Jatayu Limited compute the Economic Value Added:

Capital Structure	Equity capital ₹ 160 Lakhs Reserves and Surplus ₹ 140 lakhs 10% Debentures ₹ 400 lakhs
Cost of equity	14%
Financial Leverage	1.5 times
Income Tax Rate	30%

Solution — ICAI

9. Financial Leverage = PBIT/PBT

1.5 = PBIT / (PBIT – Interest)

1.5 = PBIT / (PBIT – 40)

1.5 (PBIT – 40) = PBIT

1.5 PBIT – 60 = PBIT

1.5 PBIT – PBIT = 60

0.5 PBIT = 60

or PBIT $= \frac{60}{0.5}$ = ₹120 lakhs

NOPAT = PBIT – Tax = ₹120 lakhs (1 – 0.30) = ₹84 lakhs.

Weighted Average Cost of Capital (WACC)

= 14% × (300 / 700) + (1 – 0.30) × (10%) × (400 / 700) = 10%

EVA = NOPAT – (WACC × Total Capital)

EVA = ₹84 lakhs – 0.10 × ₹ 700 lakhs

EVA = ₹ 14 lakhs

Practical Q10

10. RST Ltd.'s current financial year's income statement reported its net income after tax as ₹ 25,00,000. The applicable corporate income tax rate is 30%.

Following is the capital structure of RST Ltd. at the end of current financial year:

	₹
Debt (Coupon rate = 11%)	40 lakhs
Equity (Share Capital + Reserves & Surplus)	125 lakhs
Invested Capital	165 lakhs

Following data is given to estimate cost of equity capital:

Equity Beta of RST Ltd.	1.36
Risk –free rate i.e. current yield on Govt. bonds	8.5%
Average market risk premium (i.e. Excess of return on market portfolio over risk-free rate)	9%

Required:

(i) Estimate Weighted Average Cost of Capital (WACC) of RST Ltd.; and

(ii) Estimate Economic Value Added (EVA) of RST Ltd.

Solution — ICAI

10. Cost of Equity as per CAPM

$k_e = R_f + \beta$ x Market Risk Premium

= 8.5% + 1.36 x 9%

= 8.5% + 12.24% = 20.74%

Cost of Debt k_d = 11%(1 – 0.30) = 7.70%

WACC (k_o) $= k_e \times \frac{E}{E+D} + k_d \times \frac{D}{E+D}$

$= 20.74 \times \frac{125}{165} + 7.70 \times \frac{40}{165} = 15.71 + 1.87 = 17.58\%$

Taxable Income = ₹ 25,00,000/(1 - 0.30)

= ₹ 35,71,429 or ₹ 35.71 lakhs

Operating Income = Taxable Income + Interest

= ₹ 35,71,429 + ₹ 4,40,000

= ₹ 40,11,429 or ₹ 40.11 lacs

EVA = EBIT (1-Tax Rate) – WACC x Invested Capital

= ₹ 40,11,429 (1 – 0.30) – 17.58% x ₹ 1,65,00,000

= ₹ 28,08,000 – ₹ 29,00,700 = - ₹ 92,700

Practical Q11

11. Tender Ltd has earned a net profit of ₹ 15 lacs after tax at 30%. Interest cost charged by financial institutions was ₹ 10 lacs. The invested capital is ₹ 95 lacs of which 55% is debt. The company maintains a weighted average cost of capital of 13%. Required,

(a) Compute the operating income.

(b) Compute the Economic Value Added (EVA).

(c) Tender Ltd. has 6 lac equity shares outstanding. How much dividend can the company pay before the value of the entity starts declining?

Solution — ICAI

11. Taxable Income = ₹ 15 lac/(1-0.30)

= ₹ 21.43 lacs or ₹ 21,42,857

Operating Income = Taxable Income + Interest

= ₹ 21,42,857 + ₹ 10,00,000

= ₹ 31,42,857 or ₹ 31.43 lacs

EVA = EBIT (1-Tax Rate) – WACC x Invested Capital

= ₹ 31,42,857(1 – 0.30) – 13% x ₹ 95,00,000

= ₹ 22,00,000 – ₹ 12,35,000 = ₹ 9,65,000

EVA Dividend = $\frac{₹\ 9,65,000}{₹\ 6,00,000}$ = ₹ 1.6083

Practical Q12

12. The following information is given for 3 companies that are identical except for their capital structure:

	Orange	Grape	Apple
Total invested capital	1,00,000	1,00,000	1,00,000
Debt/assets ratio	0.8	0.5	0.2
Shares outstanding	6,100	8,300	10,000
Pre tax cost of debt	16%	13%	15%
Cost of equity	26%	22%	20%
Operating Income (EBIT)	25,000	25,000	25,000

The tax rate is uniform 35% in all cases.

(i) Compute the Weighted average cost of capital for each company.

(ii) Compute the Economic Valued Added (EVA) for each company.

(iii) Based on the EVA, which company would be considered for best investment? Give reasons.

(iv) If the industry PE ratio is 11x, estimate the price for the share of each company.

(v) Calculate the estimated market capitalisation for each of the Companies.

Solution — ICAI

12. (i) Working for calculation of WACC

	Orange	Grape	Apple
Total debt	80,000	50,000	20,000
Post tax Cost of debt	10.40%	8.45%	9.75%
Equity Fund	20,000	50,000	80,000

WACC

Orange:	(10.4 x 0.8) + (26 x 0.2)	=	13.52%
Grape:	(8.45 x 0.5) + (22 x 0.5)	=	15.225%
Apple:	(9.75 x 0.2) + (20 x 0.8)	=	17.95%

(ii)

	Orange	Grape	Apple
WACC	13.52	15.225	17.95
EVA [EBIT (1-T)-(WACC x Invested Capital)]	2,730	1,025	-1,700

(iii) Orange would be considered as the best investment since the EVA of the company is highest and its weighted average cost of capital is the lowest

(iv) Estimated Price of each company shares

	Orange	Grape	Apple
EBIT (₹)	25,000	25,000	25,000
Interest (₹)	12,800	6,500	3,000
Taxable Income (₹)	12,200	18,500	22,000
Tax 35% (₹)	4,270	6,475	7,700
Net Income (₹)	7,930	12,025	14,300
Shares	6,100	8,300	10,000
EPS (₹)	1.30	1.45	1.43
Stock Price (EPS x PE Ratio) (₹)	14.30	15.95	15.73

Since the three entities have different capital structures they would be exposed to different degrees of financial risk. The PE ratio should therefore be adjusted for the risk factor.

(v) Market Capitalisation

Estimated Stock Price (₹)	14.30	15.95	15.73
No. of shares	6,100	8,300	10,000
Estimated Market Cap (₹)	87,230	1,32,385	1,57,300

Practical Q13

13. Delta Ltd.'s current financial year's income statement reports its net income as ₹ 15,00,000. Delta's marginal tax rate is 40% and its interest expense for the year was ₹ 15,00,000. The company has ₹ 1,00,00,000 of invested capital, of which 60% is debt. In addition, Delta Ltd. tries to maintain a Weighted Average Cost of Capital (WACC) of 12.6%.

(i) Compute the operating income or EBIT earned by Delta Ltd. in the current year.

(ii) What is Delta Ltd.'s Economic Value Added (EVA) for the current year?

(iii) Delta Ltd. has 2,50,000 equity shares outstanding. According to the EVA you computed in (ii), how much can Delta pay in dividend per share before the value of the company would start to decrease? If Delta does not pay any dividends, what would you expect to happen to the value of the company?

Solution — ICAI

13. **(i)** Taxable income = Net Income /(1 – 0.40)

or, Taxable income = ₹ 15,00,000/(1 – 0.40) = ₹ 25,00,000

Again, taxable income = EBIT – Interest

or, EBIT = Taxable Income + Interest

= ₹ 25,00,000 + ₹ 15,00,000 = ₹ 40,00,000

(ii) EVA = EBIT (1 – T) – (WACC × Invested capital)

= ₹ 40,00,000 (1 – 0.40) – (0.126 × ₹ 1,00,00,000)

= ₹ 24,00,000 – ₹ 12,60,000 = ₹ 11,40,000

(iii) EVA Dividend = ₹ 11,40,000/2,50,000 = ₹ 4.56

If Delta Ltd. does not pay a dividend, we would expect the value of the firm to increase because it will achieve higher growth, hence a higher level of EBIT. If EBIT is higher, then all else equal, the value of the firm will increase.

Practical Q14

14. The following data pertains to XYZ Inc. engaged in software consultancy business as on 31 December 2010.

($ Million)

Income from consultancy	935.00
EBIT	180.00
Less: Interest on Loan	18.00
EBT	162.00
Tax @ 35%	56.70
	105.30

Balance Sheet

($ Million)

Liabilities	Amount	Assets		Amount
Equity Stock (10 million share @ $ 10 each)	100	Land and Building		200
		Computers & Softwares		295
Reserves & Surplus	325	Current Assets:		
Loans	180	Debtors	150	
Current Liabilities	180	Bank	100	
		Cash	40	290
	785			785

With the above information and following assumption you are required to compute

(a) Economic Value Added®

(b) Market Value Added.

Assuming that:

(i) WACC is 12%.

(ii) The share of company currently quoted at $ 50 each

Solution — ICAI

14. (a) Determination of Economic Value Added (EVA)

	$ Million
EBIT	180.00
Less: Taxes @ 35%	63.00
Net Operating Profit after Tax	117.00
Less: Cost of Capital Employed [W. No.1]	72.60
Economic Value Added	44.40

(b) Determination of Market Value Added (MVA)

	$ Million
Market value of Equity Stock [W. No. 2]	500
Equity Fund [W. No. 3]	425
Market Value Added	75

Working Notes:

(1) Total Capital Employed

Equity Stock	$ 100 Million
Reserve and Surplus	$ 325 Million
Loan	$ 180 Million
	$ 605 Million

WACC	12%
Cost of Capital employed $ 605 Million x 12%	$ 72.60 Million

(2) Market Price per equity share (A) $ 50

No. of equity share outstanding (B) 10 Million

Market value of equity stock (A) x (B) $ 500 Million

(3) Equity Fund

Equity Stock	$ 100 Million
Reserves & Surplus	$ 325 Million
	$ 425 Million

Practical Q15

15. Herbal Gyan is a small but profitable producer of beauty cosmetics using the plant Aloe Vera. This is not a high-tech business, but Herbal's earnings have averaged around ₹ 12 lakh after tax, largely on the strength of its patented beauty cream for removing the pimples.

The patent has eight years to run, and Herbal has been offered ₹ 40 lakhs for the patent rights. Herbal's assets include ₹ 20 lakhs of working capital and ₹ 80 lakhs of property, plant, and equipment. The patent is not shown on Herbal's books. Suppose Herbal's cost of capital is 15 percent. What is its Economic Value Added (EVA)?

Solution — ICAI

15. EVA = Income earned – (Cost of capital x Total Investment)

Total Investments

Particulars	Amount
Working capital	₹ 20 lakhs
Property, plant, and equipment	₹ 80 lakhs
Patent rights	₹ 40 lakhs
Total	**₹ 140 lakhs**

Cost of Capital 15%

EVA= ₹ 12 lakh – (0.15 x ₹ 140 lakhs) = ₹ 12 lakh – ₹ 21 lakh = -₹ 9 lakh

Thus, Herbal Gyan has a negative EVA of ₹ 9 lakhs.

Practical Q16

16. Constant Engineering Ltd. has developed a high tech product which has reduced the Carbon emission from the burning of the fossil fuel. The product is in high demand. The product has been patented and has a market value of ₹ 100 Crore, which is not recorded in the books. The Net Worth (NW) of Constant Engineering Ltd. is ₹ 200 Crore. Long term debt is ₹ 400 Crore. The product generates a Net Operating Profit after Tax of ₹ 84 Crore. The rate on 365 days Government bond is 10 percent per annum. Market portfolio generates a return of 12 percent per annum. The stock of the company moves in tandem with the market. Calculate Economic Value added of the company.

Solution — ICAI

16. EVA = Income Earned – (Cost of Capital x Total Investment)

Total Investments

	Amount (₹ Crore)
Net Worth	200.00
Long Term Debts	400.00
Patent Rights	100.00
Total	700.00

WACC (ko) $= k_e \times \frac{E}{E+D} + k_d \times \frac{D}{E+D}$

$= 12 \times \frac{300}{700} + 10 \times \frac{400}{700}$

= 5.14% + 5.71% = 10.85%

EVA = Profit Earned – WACC x Invested Capital

= ₹ 84 crore – 10.85% x ₹ 700 crore

= ₹ 8.05 crore

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